Yifei Cai

Postdoctoral Research Associate
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Research Interests: AI Security & Privacy; Efficient Secure Inference; Privacy-Preserving AI Systems.

Academic Appointments

Iowa State University — Department of Computer Science, Ames, IA Oct. 2025–Present
Postdoctoral Research Associate Supervisor: Prof. Chunsheng Xin

- Expand privacy-preserving machine learning research from cryptography-aware optimization toward secure and trustworthy AI systems and reusable research platforms, bringing together secure computation, modern LLMs, and real-world privacy requirements.
- Mentor Ph.D. students conducting research in AI security and privacy, providing guidance on research problem formulation, methodology development, and experimental evaluation.

Education

Old Dominion University, Norfolk, VA 2025
Ph.D. in Electrical and Computer Engineering
Advisors: Prof. Chunsheng Xin and Prof. Hongyi Wu

Dissertation: *Towards Efficient Privacy-Preserving Deep Learning: HE-Friendly Structures, Flexible Pruning, HE-Efficient Architectures, and Secure Transformer Token Drop*

Beijing Institute of Technology, Beijing, China
M.S. in Electrical Engineering 2018
B.S. in Electrical Engineering 2015

Peer-Reviewed Publications

[1] **Yifei Cai**, Zhuoran Li, Yizhou Feng, Qiao Zhang, Hongyi Wu, Danella Zhao, and Chunsheng Xin
"SecDTD: Dynamic Token Drop for Secure Transformers Inference."
IEEE European Symposium on Security and Privacy (EuroS&P), 2026. DOI

[2] **Yifei Cai**, Yizhou Feng, Qiao Zhang, Chunsheng Xin, and Hongyi Wu
"Towards Zero Rotation and Beyond: Architecting Neural Networks for Fast Secure Inference with Homomorphic Encryption."
IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2026. DOI

[3] Xiaozuo Shen, **Yifei Cai**, Rui Ning, Chunsheng Xin, and Hongyi Wu
"DF-LoGiT: Data-Free Logic-Gated Backdoor Attacks in Vision Transformers."
International Conference on Machine Learning (ICML), PMLR 306:110706–110728, 2026. Paper

[4] Zhuoran Li, Hanieh Totonchi Asl, **Yifei Cai**, Ebrahim Nouri, and Danella Zhao
"TAMI-MPC: Trusted Acceleration of Minimal-Interaction MPC for Efficient Nonlinear Inference."
Design Automation Conference (DAC), 2026. arXiv

[5] Zhuoran Li, Hanieh Totonchi Asl, Ebrahim Nouri, **Yifei Cai**, and Dan Zhao
"Silentflow: Leveraging Trusted Execution for Resource-Limited MPC via Hardware-Algorithm Co-design."
Asia and South Pacific Design Automation Conference (ASP-DAC), pp. 737–743, 2026. DOI

[6] **Yifei Cai**, Qiao Zhang, Rui Ning, Chunsheng Xin, and Hongyi Wu
"MOSAIC: A Prune-and-Assemble Approach for Efficient Model Pruning in Privacy-Preserving Deep Learning."
ACM Asia Conference on Computer and Communications Security (AsiaCCS), pp. 1034–1048, 2024. DOI

[7] Yizhou Feng, Qiao Zhang, **Yifei Cai**, Hongyi Wu, and Chunsheng Xin
"TILE: Input Structure Optimization for Neural Networks to Accelerate Secure Inference."
Annual Computer Security Applications Conference (ACSAC), pp. 1203–1216, 2024. DOI

[8] Qiao Zhang, Tao Xiang, **Yifei Cai**, Zhichao Zhao, Ning Wang, and Hongyi Wu
"Privacy-Preserving Machine Learning as a Service: Challenges and Opportunities."
IEEE Network, 37(6):214–223, 2023. DOI

[9] **Yifei Cai**, Qiao Zhang, Rui Ning, Chunsheng Xin, and Hongyi Wu
"Hunter: HE-Friendly Structured Pruning for Efficient Privacy-Preserving Deep Learning."
ACM Asia Conference on Computer and Communications Security (AsiaCCS), pp. 931–945, 2022. DOI

Preprints and Manuscripts

[P1] **Yifei Cai**, Zhuoran Li, Xiaozuo Shen, Hongyi Wu, and Chunsheng Xin
"DegreeSpar: Structured Degree Sparsity for Efficient Secure Transformer Inference."
arXiv preprint, under review, 2026. arXiv

[P2] Xiaozuo Shen, **Yifei Cai**, Tian Tan, Rui Ning, Chunsheng Xin, and Hongyi Wu
"When Trees Are Not Enough: Learning Mixed-Topology
Feature Graphs with Adaptive Graph Sparse Autoencoders."
arXiv preprint, under review, 2026. arXiv

Selected Research Contributions

Cryptography-Aware Model Optimization. Pioneered HE-aware structured pruning with *Hunter* (AsiaCCS 2022), helping establish cryptography-aware optimization of linear computation in HE-based privacy-preserving machine learning. Hunter has served as a reference baseline for subsequent research, including *MOSAIC* and HE-efficient neural architecture optimization.

Efficient Secure Transformer Inference. Developed *SecDTD*, a secure dynamic token-dropping approach achieving up to $4.47\times$ end-to-end acceleration for secure Transformer inference. Its oblivious median-selection protocol, *OMSel*, achieves up to $16.9\times$ speedup over Bitonic Sort, addressing a key latency bottleneck in secure token selection under MPC.

Selective Privacy Protection for Practical AI (Ongoing). Investigating practical privacy-preserving LLM inference through selective protection of sensitive information and the separation of plaintext and encrypted computation. This research explores practical trade-offs among privacy protection, model utility, and computational efficiency in real-world AI applications.

Teaching Experience

Old Dominion University, Norfolk, VA Jan. 2023–Aug. 2025
Tutor, School of Cybersecurity

- Provided supplemental instruction to undergraduate and graduate students, reinforcing classroom teaching and helping students develop a deeper understanding of cybersecurity concepts.
- Guided students through hands-on cybersecurity labs and attack-and-defense exercises, helping them apply security concepts to realistic scenarios and develop practical cybersecurity skills.
- Supported CYSE 301 (undergraduate); CYSE 601, CYSE 602, and CYSE 695 (graduate).

Proposal Development

- **NSF Cybersecurity Research Proposal.** Developed technical sections and contributed to the preparation of an NSF research proposal on cryptography-aware model compression and efficient private inference, building on prior research in HE-friendly pruning, neural architecture design, and secure Transformers.
- **Interdisciplinary Privacy-Preserving AI Platform.** Developing technical concepts for an open research platform integrating automated secure-model conversion, selective data protection, and distributed edge-cloud computing to support privacy-sensitive AI applications.

Professional Service

IEEE Transactions Manuscript Reviewing 2023–Present
Reviewed multiple manuscripts for *IEEE Transactions on Computers*, *IEEE Transactions on Cloud Computing*, *IEEE Transactions on Services Computing*, and *IEEE Transactions on Mobile Computing*.

Last updated: October 2026.